Interpolation With Function Space Representation of Membership Functions

Yeung Yam, *Senior Member, IEEE*, Man Lung Wong, and Péter Baranyi

***Abstract*—This paper generalizes a previous Cartesian approach for interpolating fuzzy rules comprised of membership functions with finite number of characteristic points. Instead of representing membership functions as points in Cartesian spaces, they now become elements in the space of square, integrable function. Interpolation is thus conducted between the antecedent and consequent function spaces. The generalized representation allows an extended class of membership functions satisfying two monotonicity conditions to be accommodated in the interpolation process. They include the popular bell-shaped membership functions, which were not possible before with the Cartesian representation. The work also extends the similarity triangle-based interpolation technique from the previous Cartesian representation to the new representation. Ensuing issues on computational complexity and nonunique conclusion are discussed. Other concepts such as spanning set and extensibility functions are also presented under the generalized framework. Examples to illustrate the extended approach and to compare with the Cartesian approach are given.**

I. Introduction

FUZZY approximation often times relies on using dense rule bases and large numbers of antecedent variables and linguistic terms to attain high precision. Such practice, however, often leads to undesirable computational delay, storage and retrieval problem. One approach to alleviate this "curse of dimensionality" situation is to adopt sparse fuzzy rules, and then apply interpolation to extract conclusion for observation falling into regions not covered by the antecedents [1], [2]. A number of interpolation techniques have been proposed in this regards, ranging from point-by-point extraction [1], to solid cutting techniques [3], and to the conservation of relative fuzziness [4], [5]. It is worth noting that fuzzy interpolation is applicable not only for sparse rules, but also to dense ones during their buildup phase.

Recently, an approach to represent membership functions as points in Cartesian space was introduced [6], [7]. Under this representation, a fuzzy rule base can be viewed as mappings between finite dimensional antecedent and consequent spaces, and the interpolation problem thus becomes searching for an image in the consequent space for the given antecedent observation. Various interpolation techniques can be expressed as matrix multiplication operators between the two spaces [8]. Moreover, the representation has the advantage of allowing "well-defined" region for membership functions to be characterized and incorporated in the interpolation process [9]. Application to extract sparse rules from a given dense set has also been conducted [10].

The Cartesian representation is applicable to membership functions with finite number of characteristic points only, such as triangular and trapezoidal ones. Application to smooth membership functions not comprising of piecewise linear segments, e.g., bell-shaped membership functions, is not possible. Even membership functions with a finite but large number of characteristic points pose a problem, as they involve Cartesian spaces of high dimensions. To improve the situation, a preliminary idea to represent membership functions as elements in function space was proposed in [11]. Under the framework, membership functions satisfying certain monotonicity conditions are represented as elements of the function space of real, square integrable functions [12] within the interval [0,2].

This work constitutes an detailed expansion of the preliminary idea in [11] and its application to the interpolation problem. In particular, we focus on the similarity triangle interpolation approach of [7] and compare the ensuing conditions and procedures under the Cartesian and function space representations. It is shown that computational complexity in the new representation can be efficiently tackled with the defined basis functions and inner products. Other concepts such as the spanning set and extensibility function introduced for the Cartesian framework are extended as well. Numerical examples to illustrate the function space approach and to compare with the Cartesian results are also given.

This paper is structured as follows. Sections II and III give a brief review of the Cartesian representation and its application to the interpolation problem. The similarity triangle interpolation method is especially highlighted. Section IV introduces the extended representation in function space. Section V presents the similarity triangle interpolation method under the new representation. An example involving various types of membership functions is given. Section VI presents the basis functions and their inner products to facilitate efficient computation. The incorporation of additional objective to uniquely determine a conclusion is also formulated. Section VII extends the concepts of spanning set and extensibility functions and their applications to the function space, and finally, Section VIII presents the conclusions.

II. Cartesian Representation

The Cartesian representation calls for representing membership function as distinct point in Cartesian space. Specifically,

Manuscript received August 11, 2003; revised September 16, 2005. This work was supported in part by RGC Direct Grant 2050234.

Y. Yam and M. L. Wong are with the Intelligent Control Systems Laboratory, Department of Automation and Computer-Aided Engineering, The Chinese University of Hong Kong, Shatin, N.T., Hong Kong (e-mail: yyam@acae.cuhk.edu.hk; mlwong@acae.cuhk.edu.hk).

P. Baranyi is with the Department of Telecommunications and Media Informatics, Budapest University of Economic and Technology and Computer and Automation Research Institute of the Hungarian Academy of Sciences, Budapest H-1111, Hungary (e-mail: baranyi@ttt-202.ttt.bme.hu).

Digital Object Identifier 10.1109/TFUZZ.2006.876332

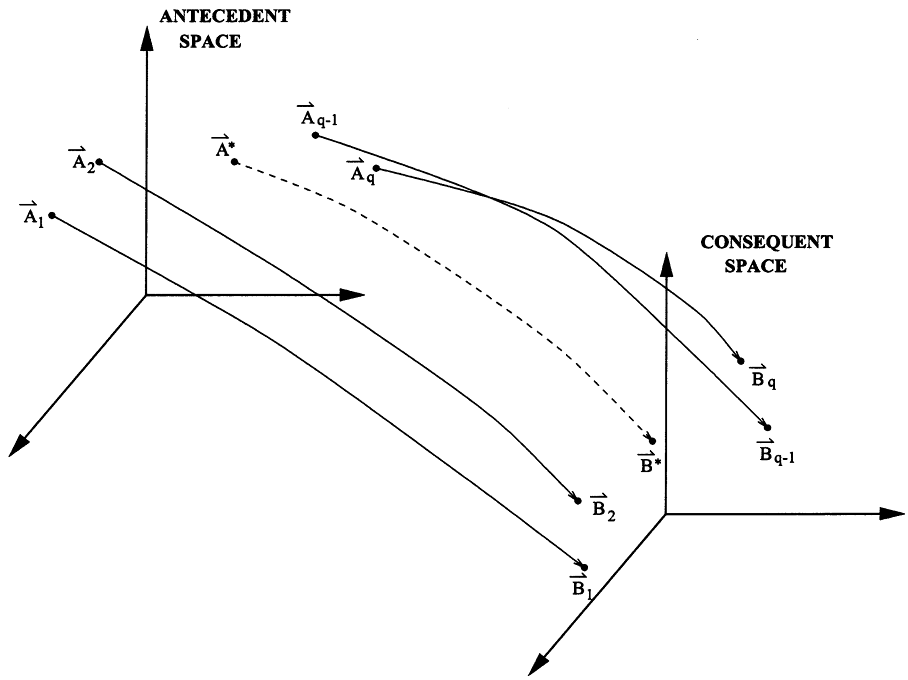

Fig. 1. Interpolation as mapping between antecedent and consequent spaces.

let F be a membership function of fuzzy variable f and is comprised of n characteristic points, then F can be represented as a point $\vec{F}$ in the n-dimensional Cartesian space $\mathcal{R}^n$. The coordinates of the point $\vec{F}$ are

$$\vec{F} = \begin{bmatrix} f^{(1)} \\ f^{(2)} \\ \vdots \\ f^{(n)} \end{bmatrix} \in \mathcal{R}^n \tag{1}$$

where $f^{(i)}$ is the value of f at the ith characteristic point. With this representation, a fuzzy set of q rules: If $A_i \Rightarrow B_i$, $i = 1, \ldots, q$, can be viewed as mappings between points $\vec{A}_i$ in the antecedent space $\mathcal{R}^n$ to points $\vec{B}_i$ in the consequent space $\mathcal{R}^m$ as in Fig. 1. Here, n and m are the number of characteristic points for membership functions A_i and B_i, respectively. The interpolation problem thus becomes the search for a proper image $\vec{B}^*$ in $\mathcal{R}^m$ for the given observation $\vec{A}^*$ in $\mathcal{R}^n$.

One shortcoming in some of the earlier interpolation techniques, e.g., [1], is that they may result in conclusions of abnormal membership function, in which case more processing is needed to obtain interpretable results. The Cartesian representation, on the other hand, enables an efficient characterization of the well-defined region for membership functions and allows the abnormality problem to be tackled during the interpolation process. Take the case of $\mathcal{R}^3$, which corresponds to membership functions with three characteristic points. By definition, a membership function F is well-defined if its $(i+1)$th characteristic point does not occur before the ith one, i.e., $f^{(3)} \geq f^{(2)} \geq f^{(1)}$. With (x, y, z) denoting the coordinates in $\mathcal{R}^3$, the well-defined region can therefore be characterized by

$$z \geq y \geq x. \tag{2}$$

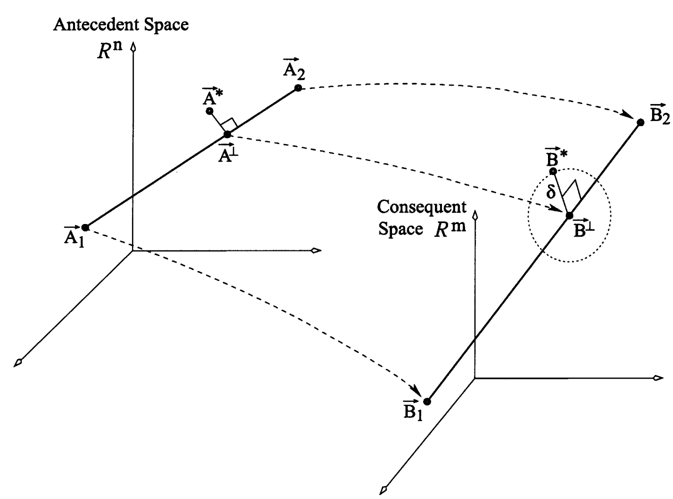

Fig. 2. Similarity triangle interpolation approach.

The interpolation problem is then to locate an image in consequent space within the well-defined region. A technique to guarantee well-defined conclusions using (2) is given in [9].

III. Similarity Triangle Interpolation in $\mathcal{R}^n$

The similarity triangle interpolation method was introduced in [6] and [7] as an application of the Cartesian representation. To illustrate the main idea of the method, we consider a simple problem of two sparse rules: $A_i \rightarrow B_i, i = 1, 2$, and a given observation A^*. Fig. 2 depicts the Cartesian representation of the problem. The similarity triangle method calls for determining the conclusion B^* by requiring that the triangles $A_1A^*A_2$ and $B_1B^*B_2$ be similar to each other. The method first generates an intermediate point $A^\perp$ on the line A_1A_2 such that $A^*A^\perp$ is

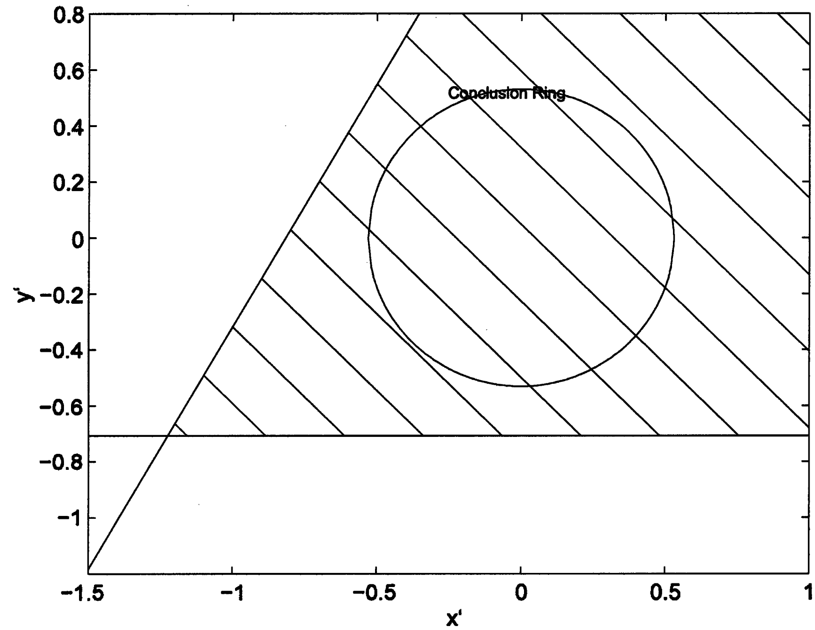

Fig. 3. Conclusion ring and well-defined (shaded) region for Example 1 with Cartesian interpolation.

perpendicular to A_1A_2. The image of $A^\perp$, $B^\perp$, can then be obtained by cutting the line B_1B_2 at the same ratio as $A^\perp$ cuts A_1A_2. This is termed characteristic interpolation; no additional information outside the fuzzy rules would be required. In contrast, the remaining part to determine the conclusion B^* protruding from $B^\perp$ orthogonal to the line B_1B_2 is called orthogonal interpolation, which would require additional information from outside, heuristic or otherwise. In vector notation, the previous procedures are expressed as follows.

1) Compute the unit vector $\vec{a}_{12}$ in the direction of $(\vec{A}_2 - \vec{A}_1)$

$$\vec{a}_{12} = \frac{(\vec{A}_2 - \vec{A}_1)}{|\vec{A}_2 - \vec{A}_1|_n}. \tag{3}$$

2) Determine $\vec{A}^\perp$

$$\vec{A}^\perp = \vec{A}_1 + \left[(\vec{A}^* - \vec{A}_1)\cdot \vec{a}_{12}\right]\vec{a}_{12}. \tag{4}$$

3) Compute the ratios γ_1 and γ_2

$$\gamma_1 = 1 - \gamma_2 \quad \gamma_2 = \frac{|\vec{A}_1 - \vec{A}^\perp|}{|\vec{A}_1 - \vec{A}_2|_n}. \tag{5}$$

4) Obtain $\vec{B}^\perp$

$$\vec{B}^\perp = \gamma_1\vec{B}_1 + \gamma_2\vec{B}_2. \tag{6}$$

5) Obtain conclusion set $\vec{B}^*$ based on the similarity triangle condition

$$\vec{B}^* = \vec{B}^\perp + \mathcal{B}_{m-1} \tag{7}$$

where $\mathcal{B}_{m-1}$ denotes a $(m-1)$-dimensional shell of radius δ orthogonal to $(\vec{B}_1 - \vec{B}_2)$. The value δ is given by the scaling equation

$$\delta = |\vec{B}^* - \vec{B}^\perp|_m = |\vec{A}^* - \vec{A}^\perp|_n \frac{|\vec{B}_2 - \vec{B}_1|_m}{|\vec{A}_2 - \vec{A}_1|_n}. \tag{8}$$

Here, $|\cdot|_n$ and $|\cdot|_m$ are the Euclidean norms in $\mathcal{R}^n$ and $\mathcal{R}^m$, respectively. Steps 1)–3) are to be performed on the antecedent space, and steps 4) and 5) on the consequent space. Also, depending on the value m, the conclusion may not be unique. For $m = 3$, say, the set of possible conclusion would be a ring of radius δ centering at $B^\perp$ on a perpendicular plane to the line B_1B_2, as depicted in Fig. 2. At this point, the subset of $\vec{B}^*$ lying inside the well-defined region would constitute valid conclusions.

Example 1: Consider the interpolation problem with the following membership functions:

$$\vec{A}_1 = \begin{bmatrix}0\\1\\2\end{bmatrix} \vec{A}_2 = \begin{bmatrix}8\\9\\10\end{bmatrix} \vec{B}_1 = \begin{bmatrix}1\\2\\3\end{bmatrix}$$
$$\vec{B}_2 = \begin{bmatrix}7\\8\\9\end{bmatrix} \vec{A}^* = \begin{bmatrix}4.5\\5.0\\5.5\end{bmatrix}.$$

The membership functions all have three characteristic points, $m = n = 3$. Carrying out the aforementioned procedures, we have $\vec{A}^\perp = [4\ 5\ 6]^T$, $\gamma_1 = 0.5$, $\gamma_2 = 0.5$, and accordingly, $\vec{b}^\perp = [4\ 5\ 6]^T$ and $\delta = 0.53$. Fig. 3 shows the conclusion ring from (7) superimposed on the well-defined region (shaded area). The coordinate frame in the figure has been translated to have $\vec{B}^\perp$ as origin, and rotated to have the conclusion ring flat on the resulting xy plane. The conclusion ring in this case lies completely within the well-defined region. Any choice on the conclusion ring will produce a valid conclusion. Fig. 4 plots a few possible conclusions distributed over the ring. One may at this point incorporates additional objective(s) to narrow the choices of possible conclusions, or to uniquely determine one.

IV. Representation in Function Space $L_2[0, 2]$

The Cartesian representation is best applicable in situation when the number of characteristic points is low. The approach is cumbersome if the membership functions contain a large

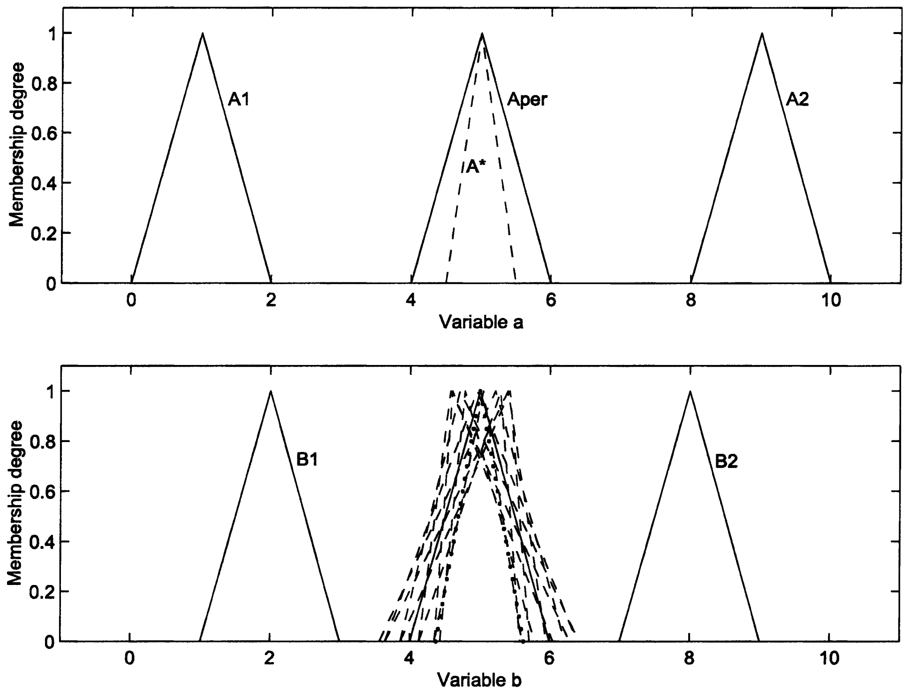

Fig. 4. Possible conclusions for Example 1 with Cartesian interpolation.

number of characteristic points, in which case the corresponding Cartesian spaces will be of high dimensions. Even more so, application to membership functions which are not piecewise linear, such as the popular bell-shaped membership functions, is not obvious. In this regards, a function space representation was proposed in [11] for an extended class of membership functions over those with finite number of characteristic points only. As depicted in Fig. 5(a), membership function $A(x)$ is divided at the point where the membership degree is equal to 1. Let $A_L(x)$ and $A_R(x)$ denote, respectively, the left- and right-hand sides of the membership function. Assume the following.

- Condition C_1: $A_L(x)$ is nondecreasing going from 0 to 1.
- Condition C_2: $A_R(x)$ is nonincreasing going from 1 to 0.

Conditions C_1 and C_2 imply the existence of the inverses of $A_L(x)$ and $A_R(x)$. One can take the inverse of $A_L(x)$ to form $A_L^{-1}(\mu)$ for $0 \leq \mu < 1$, and the inverse of $A_R(x)$ to form $A_R^{-1}(2-\mu)$ for $1 \leq \mu \leq 2$. A function $g_A(\mu)$ to represent $A(x)$ in $L_2[0,2]$ can then be constructed

$$g_A(\mu) = \begin{cases} A_L^{-1}(\mu), & 0 \leq \mu < 1 \\ A_R^{-1}(2-\mu), & 1 \leq \mu \leq 2 \end{cases} \tag{9}$$

as shown in Fig. 5(b). In the special case where $A_L(x)$ has constant value μ_c within $x_1 \leq x \leq x_2$, we let $A_L^{-1}(\mu_c) \equiv x_2$, and where $A_R(x)$ has constant value μ_c within $x_1 \leq x \leq x_2$, we let $A_R^{-1}(2-\mu_c) \equiv x_2$. The resulting $g_A(\mu)$ is a discontinuous function.

The inner product and the norm adopted for $L_2[0,2]$ are

$$\langle g_1, g_2 \rangle = \frac{1}{2} \int_0^2 g_1(\mu) g_2(\mu) d\mu \tag{10}$$

$$|g_1| = \sqrt{\langle g_1, g_1 \rangle}. \tag{11}$$

Equation (10) includes an optional scaling factor of 1/2 to account for the fact that integration is from 0 to 2. The monotonicity conditions C_1 and C_2 define the representable class of membership function. As depicted in Fig. 6, $L_2[0,2]$-representation is possible for membership functions of the type (a)–(d), which are the more popular ones found in literatures. Membership functions (e) and (f), however, are not representable as they do not satisfy the monotonicity conditions C_1 and C_2. Fig. 7 shows the respective representations $g_A(\mu)$ for the membership functions (a)–(d). Function (b) has a membership degree of 1 for $6 \leq x \leq 7$. The resulting $g_A(\mu)$ is

$$g_A(\mu) = \begin{cases} 5+\mu, & 0 \leq \mu < 1 \\ 6+\mu, & 1 \leq \mu \leq 2 \end{cases} \tag{12}$$

which is discontinuous as mentioned above. Moreover, case (d) illustrates the fact that the region for well-defined membership function in $L_2[0,2]$ space takes the form of

$$\frac{dg_A(\mu)}{d\mu} \geq 0. \tag{13}$$

Equation (13) constitutes the generalized version of (2) in the $L_2[0,2]$ space.

V. Similarity Triangle Interpolation in $L_2[0,2]$

A detail study into applying the $L_2[0,2]$ representation for fuzzy interpolation is given in this Section. Consider again the sparse rules $A_i \rightarrow B_i$, $i = 1, 2$, with observation A^*. Let $g_{A_1}, g_{A_2}, g_{A^*}, g_{B_1}, g_{B_2}$ be the respective representations of the membership functions in $L_2[0,2]$. Procedures for the similarity triangle interpolation can be expressed as follows.

1) Find the function $g_{a_{12}}$

$$g_{a_{12}} = \frac{(g_{A_2} - g_{A_1})}{|g_{A_2} - g_{A_1}|}. \tag{14}$$

2) Determine $g_{A^\perp}$

$$g_{A^\perp} = g_{A_1} + \langle (g_{A^*} - g_{A_1}), g_{a_{12}} \rangle g_{a_{12}}. \tag{15}$$

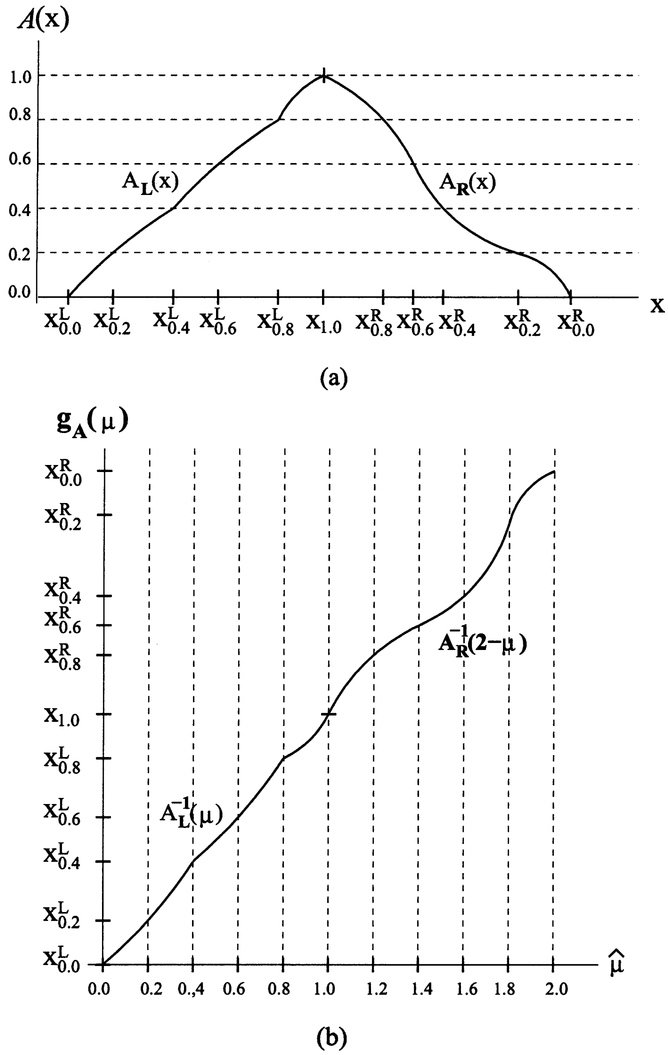

Fig. 5. Membership function and its $L_2[0,2]$ representation. (a) Membership function. (b) Corresponding representation in $L^2[0,2]$.

3) Compute the ratios τ_1 and τ_2

$$\tau_1 = 1-\tau_2 \quad \tau_2 = \frac{|g_{A_1} - g_{A^{\perp}}|}{|g_{A_1} - g_{A_2}|}. \tag{16}$$

4) Obtain $g_{B^{\perp}}$

$$g_{B^{\perp}} = \tau_1 g_{B_1} + \tau_2 g_{B_2}. \tag{17}$$

5) The conclusion set g_{B^*} satisfying the similarity property is then

$$g_{B^*} = g_{B^{\perp}} + \mathcal{G}_{\mathcal{B}} \tag{18}$$

where $\mathcal{G}_{\mathcal{B}}$ denotes the set of function orthogonal to $(g_{B_2} - g_{B_1})$ with a norm value of Δ, and Δ satisfies the scaling equation for similarity triangle

$$\Delta = |g_{A^*}(\mu) - g_{A^{\perp}}(\mu)| \frac{|g_{B_2}(\mu) - g_{B_1}(\mu)|}{|g_{A_2}(\mu) - g_{A_1}(\mu)|}. \tag{19}$$

Specifically, let $\Upsilon(\mu)$ be an element in $\mathcal{G}_{\mathcal{B}}$, $\Upsilon(\mu) = (g_{B^*} - g_{B^{\perp}})$. Thus, $\Upsilon(\mu)$ has norm Δ

$$\frac{1}{2}\int_0^2 \Upsilon^2(\mu)d\mu = \Delta^2 \tag{20}$$

and also satisfies the orthogonal condition $\langle \Upsilon(\mu), (g_{B_2} - g_{B_1})\rangle = 0$, or

$$\frac{1}{2}\int_0^2 \Upsilon(\mu)\left(g_{B_2}(\mu) - g_{B_1}(\mu)\right)d\mu = 0. \tag{21}$$

Moreover, the conclusion g_{B^*} should be well-defined condition. From (13), this means

$$\frac{dg_{B^*}(\mu)}{d\mu} \geq 0 \tag{22}$$

or, by (18)

$$\frac{d\Upsilon(\mu)}{d\mu} \geq -\frac{dg_{B^{\perp}}(\mu)}{d\mu}. \tag{23}$$

Equations (20), (21), and (23) give a complete characterization of the interpolated conclusion under $L_2[0,2]$ representation. Depending on the number of parameters defining $\Upsilon(\mu)$, the procedures may result in nonunique conclusions as in the Cartesian representation case. More specifically, if $\Upsilon(\mu)$ is characterized by m parameters, the norm and orthogonal conditions (20) and (21) will produce two constraining equations and reduce the number of free parameters to $(m-2)$ in the conclusion set, upon which the well-defined condition (23) further adds an inequality constraint to the possible degree of freedoms.

To proceed further, we consider a few special forms of $\Upsilon(\mu)$. First, let $\Upsilon(\mu)$ be triangular membership function. In $L_2[0,2]$, this means that $\Upsilon(\mu)$ is of the form

$$\Upsilon_T(\mu) = \begin{cases} a + b\mu, & 0 \leq \mu < 1 \\ a + b + c(\mu - 1), & 1 \leq \mu \leq 2 \end{cases}. \tag{24}$$

The well-defined condition (23) yields in this case

$$\begin{cases} b \geq -\frac{dg_{B^{\perp}}}{d\mu}, & 0 \leq \mu < 1 \\ c \geq -\frac{dg_{B^{\perp}}}{d\mu}, & 1 \leq \mu \leq 2 \end{cases}. \tag{25}$$

Another choice of $\Upsilon(\mu)$ is the curved membership functions of the form

$$\Upsilon_C(\mu) = a + b\mu + c\mu^2, \quad \text{for } 0 \leq \mu \leq 2 \tag{26}$$

Fig. 8 shows three examples of $\Upsilon_C(\mu)$ for different values of a, b, and c. The well-defined condition in this case is

$$b + 2c\mu \geq -\frac{dg_{B^{\perp}}}{d\mu}, \qquad 0 \leq \mu \leq 2. \tag{27}$$

Yet another possible choice of $\Upsilon(\mu)$ is the bell-shaped membership functions of the form:

$$\Upsilon_B(\mu) = \begin{cases} r - \frac{cos^{-1}(2\mu-1)}{p\pi}, & 0 \leq \mu < 1 \\ r + \frac{1}{p}\left(1 - \frac{cos^{-1}(2(\mu-1)-1)}{\pi}\right), & 1 \leq \mu \leq 2 \end{cases} \tag{28}$$

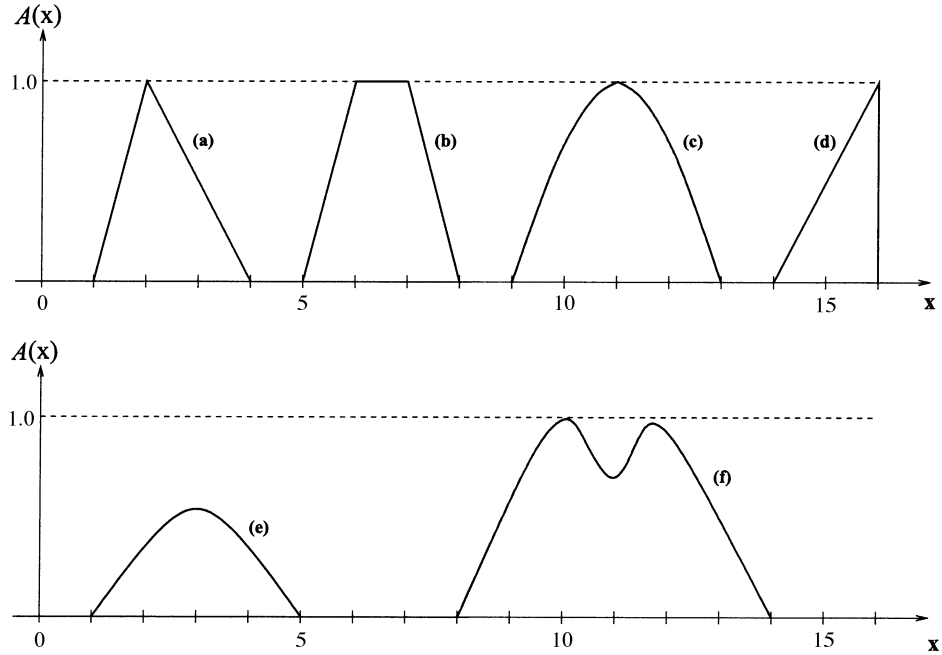

Fig. 6. Selected membership functions (a)–(f).

where the output of the function cos^{-1} is confined between 0 and π radians. Equation (28) is the $L_2[0,2]$ representation of the following bell-shaped membership function from [13]:

$$F(f) = \frac{[1+cos\,(p\pi(f-r))]}{2} \quad f \in \left[r-\frac{1}{p}, r+\frac{1}{p}\right] \tag{29}$$

with r and $2/p$ being, respectively, the central location and basewidth of $\Upsilon_B(\mu)$. In this case, the well-defined condition (23) is always satisfied.

An example is now given to illustrate the $L_2[0,2]$ interpolation.

Example 2: This example takes the triangular membership functions of Example 1, converts them to function space representation, and then conducts $L_2[0,2]$ similarity triangle interpolation. The procedures above yield $g_{A^\perp}(\mu) = 4+\mu$ and $g_{B^\perp}(\mu) = 4+\mu$ for $0 \leq \mu \leq 2$, which are actually the $L_2[0,2]$-form of $A^\perp$ and $B^\perp$ in Example 1, and also $\Delta = 0.2165$. We now consider the three different types of membership functions for the orthogonal component $\Upsilon(\mu)$.

Case(a) Triangular Type: With $\Upsilon_T(\mu)$, conditions (20) and (21) yield

$$a = -\left(\frac{3}{4}b + \frac{1}{4}c\right) \tag{30}$$

$$b^2 + 1.2bc + c^2 = 9.6\Delta^2. \tag{31}$$

Also, $g_{B^\perp}(\mu) = 4+\mu$, and hence $(dg_{B^\perp}/d\mu) = 1$, the well-defined condition (25) yields

$$\begin{cases} b \geq -1 \\ c \geq -1 \end{cases}. \tag{32}$$

Fig. 9 shows the feasible solutions in the form of an ellipse superimposed on the well-defined region (shaded) on the (c,b) plane upon proper coordinate transformation. As the ellipse lies completely inside the well-defined region, any point on it would correspond to a valid conclusion. Note that both $g_{B^\perp}$ and the orthogonal components $\Upsilon(\mu)$, and hence the interpolated conclusion, are all in this case triangular membership functions. Fig. 10 shows the resulting conclusions corresponding to the few selected points (D)–(G) of Fig. 9. Compared to Figs. 3 and 4, the possible conclusions obtained in $L_2[0,2]$ interpolation are quite similar to those from Cartesian interpolation.

Case(b) Curved Type: For curved membership function $\Upsilon_C(\mu)$, conditions (20) and (21) become

$$a = -\left(b + \frac{4}{3}c\right) \tag{33}$$

$$b^2 + 4bc + \frac{64}{15}c^2 = 3\Delta^2. \tag{34}$$

The well-defined condition (27) yields

$$b + 2c\mu \geq -1 \quad 0 \leq \mu \leq 2 \tag{35}$$

which leads to

$$\begin{cases} b \geq -1 \\ b + 4c \geq -1 \end{cases} \tag{36}$$

calculating at the values $\mu = 0$ and $\mu = 2$. Fig. 11 shows the ellipse satisfying the norm and orthogonal conditions (33) and (34) superimposed on the well-defined region (shaded) in the (c,b) plane. In this case, part of the ellipse lies outside the well-defined region and does not give rise to valid conclusions. Also, we have now $g_{B^\perp}$ a triangular membership function and the orthogonal components $\Upsilon(\mu)$ a curved membership function. The interpolated conclusion is the sum of the two. Fig. 12 shows the few conclusions corresponding to the points (D)–(G) in the well-defined region of Fig. 11.

Case(c) Bell-Shaped Type: With bell-shaped function $\Upsilon_B(\mu)$, the orthogonal condition (21) yields

$$r = 0. \tag{37}$$

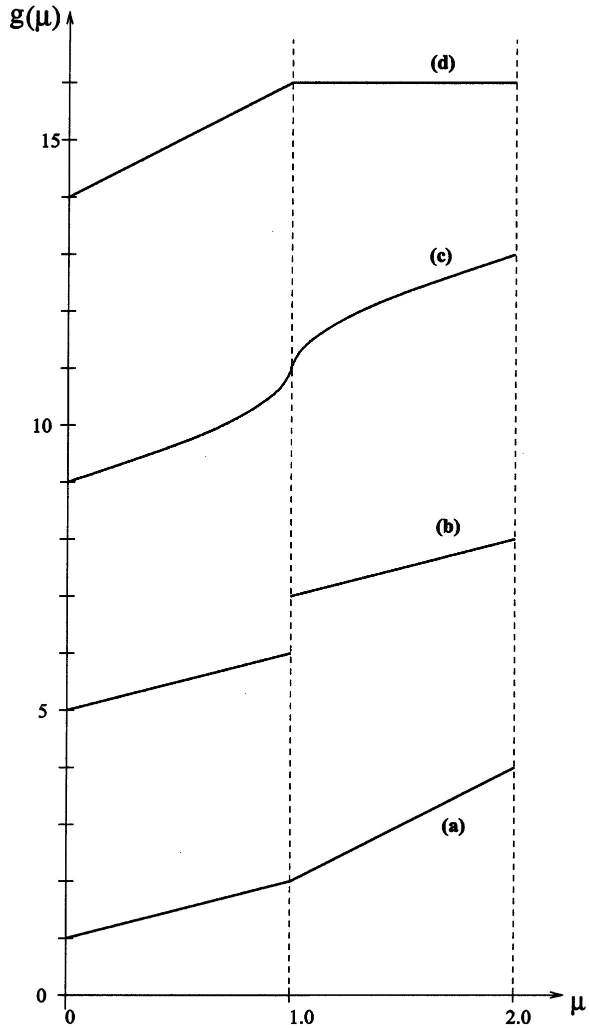

Fig. 7. $L_2[0,2]$ representations for membership functions (a)–(d) of Fig. 6.

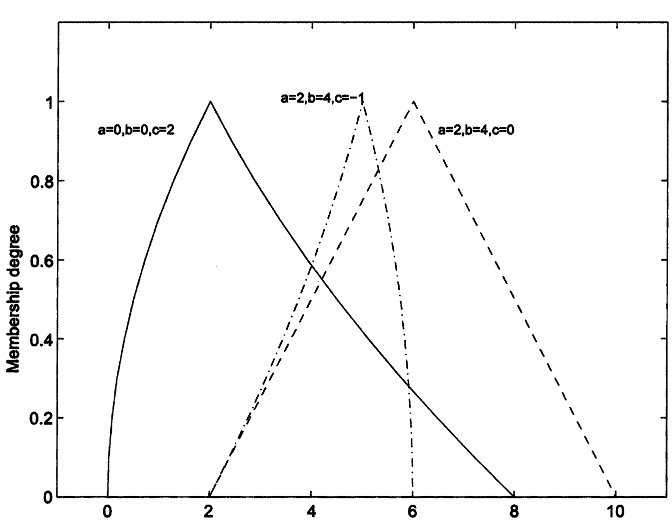

Fig. 8. Examples of curved type membership functions $\Upsilon_C(\mu)$ for different parameter values.

The norm condition (20) becomes

$$r^2 + \left(\frac{1}{p}\right)^2 \left(\frac{\pi^2 - 4}{2\pi^2}\right) = \Delta^2 \tag{38}$$

which implies $p = \pm 2.52$. Noting that $2/p$ is the basewidth of $\Upsilon_B(\mu)$ and so must be positive, the conclusion is, hence

$$\begin{aligned} g_{B^*} &= \Upsilon_B(\mu) + g_{B^\perp} \\ &= \begin{cases} (4+\mu) - \frac{cos^{-1}(2\mu-1)}{2.52\pi}, & 0 \le \mu < 1 \\ (4+\mu) \\ \quad + \frac{1}{2.52}\left(1 - \frac{cos^{-1}(2\mu-1)}{\pi}\right), & 1 \le \mu \le 2 \end{cases} \end{aligned} \tag{39}$$

In this case, $g_{B^\perp}$ is a triangular membership function and the orthogonal components $\Upsilon(\mu)$ is a bell-shaped one. The interpolation result is shown in Fig. 13.

VI. Computation Complexity and Unique Conclusion

This section addresses two specific issues inherent to the proposed work. The first is the potential shortcoming of computational complexity. Even in the simplest case, inner product is needed to achieve interpolation under $L_2[0,2]$ representation. This in turns requires the computation of integrals, which raises the concern of critical response in generating interpolated result. In this regards, we introduce the concept of basis functions and their inner products to facilitate an efficient computation. Take the example of the curved membership function $\Upsilon_C(\mu)$ of (26) characterized by the three parameters of a, b, and c. We can write

$$\Upsilon_C(\mu) = ah_1^C(\mu) + bh_2^C(\mu) + ch_3^C(\mu) \tag{40}$$

where $h_1^C(\mu) = 1$, $h_2^C(\mu) = \mu$, and $h_3^C(\mu) = \mu^2$, $0 \le \mu \le 2$, are the three corresponding basis functions. Moreover, let the quantity $(g_{B_2} - g_{B_1})$ be also expressed in the basis functions $h_i^C(\mu)$'s

$$(g_{B_2} - g_{B_1}) = D_1 h_1^C + D_2 h_2^C + D_3 h_3^C. \tag{41}$$

Conditions (20) and (21) then become

$$\begin{aligned} & a^2 \langle h_1^C, h_1^C \rangle + b^2 \langle h_2^C, h_2^C \rangle + c^2 \langle h_3^C, h_3^C \rangle \\ & \quad + 2ab \langle h_1^C, h_2^C \rangle + 2ac \langle h_1^C, h_3^C \rangle \\ & \quad + 2bc \langle h_2^C, h_3^C \rangle = \Delta^2 \end{aligned} \tag{42}$$

$$\begin{aligned} & aD_1 \langle h_1^C, h_1^C \rangle + bD_2 \langle h_2^C, h_2^C \rangle + cD_3 \langle h_3^C, h_3^C \rangle \\ & \quad + (aD_2 + bD_1) \langle h_1^C, h_2^C \rangle \\ & \quad + (aD_3 + cD_1) \langle h_1^C, h_3^C \rangle \\ & \quad + (bD_3 + cD_2) \langle h_2^C, h_3^C \rangle = 0. \end{aligned} \tag{43}$$

The inner products $\langle h_i^C, h_j^C \rangle$, $i, j = 1, 2, 3$, can be computed beforehand. In this case, we have

$$\langle h_1^C, h_1^C \rangle = 1 \quad \langle h_2^C, h_2^C \rangle = \frac{4}{3} \quad \langle h_3^C, h_3^C \rangle = \frac{16}{5} \tag{44}$$

$$\langle h_1^C, h_2^C \rangle = 1 \quad \langle h_1^C, h_3^C \rangle = \frac{4}{3} \quad \langle h_2^C, h_3^C \rangle = 2. \tag{45}$$

Conditions (42) and (43) require only substitution of the relevant values D_1, D_2, and D_3 to yield the equalities on a, b, and c dictating the set of possible conclusions.

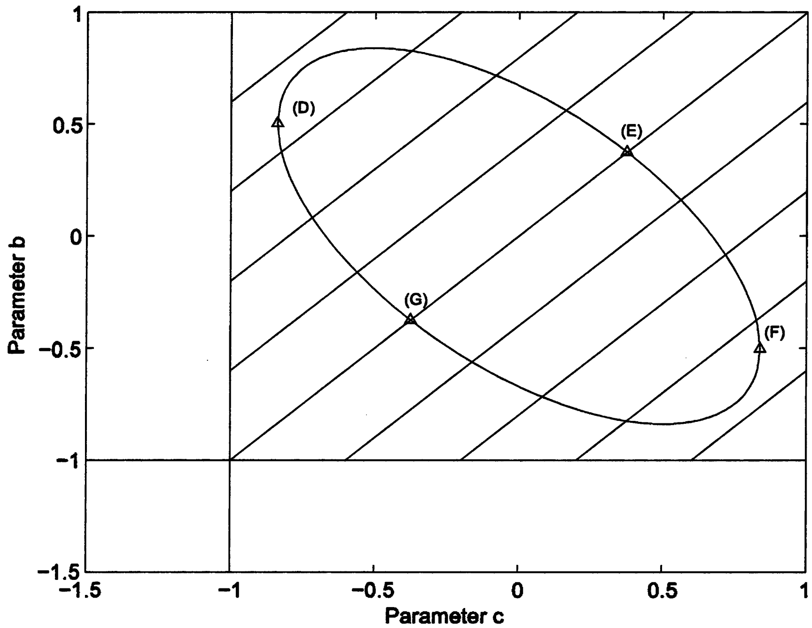

Fig. 9. Conclusions and well-defined (shaded) region for Example 2 with triangular type orthogonal component.

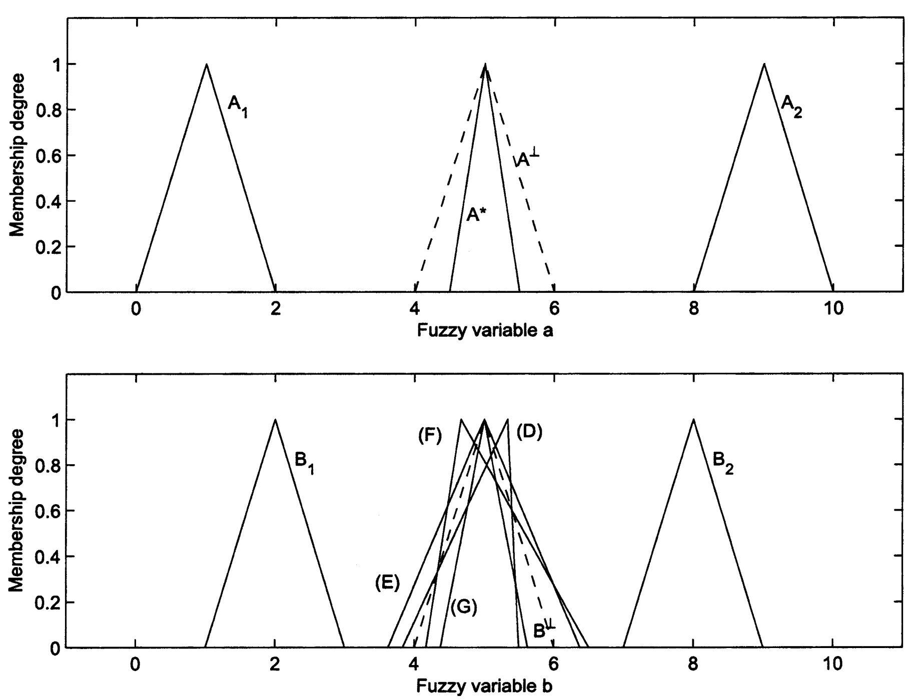

Fig. 10. $L_2[0,2]$-interpolation for Example 2 with triangular type orthogonal component.

The same can be applied to cases where $\Upsilon(\mu)$ is characterized by triangular or bell-shaped membership functions. For triangular type, we have

$$\Upsilon_T(\mu) = ah_1^T(\mu) + bh_2^T(\mu) + ch_3^T(\mu) \tag{46}$$

where the $h_i^T(\mu)$'s are given by

$$h_1^T(\mu) = 1, \qquad 0 \le \mu \le 2$$

$$h_2^T(\mu) = \begin{cases} \mu, & 0 \le \mu < 1 \\ 1, & 1 \le \mu \le 2 \end{cases}$$

$$h_3^T(\mu) = \begin{cases} 0, & 0 \le \mu < 1 \\ \mu - 1, & 1 \le \mu \le 2 \end{cases}.$$

The resulting conditions (20) and (21) are the same as (42) and (43), except that the inner products are given by $\langle h_i^T, h_j^T \rangle$, with

$$\langle h_1^T, h_1^T \rangle = 1 \quad \langle h_2^T, h_2^T \rangle = \frac{2}{3} \quad \langle h_3^T, h_3^T \rangle = \frac{1}{6} \tag{47}$$

$$\langle h_1^T, h_2^T \rangle = \frac{3}{4} \quad \langle h_1^T, h_3^T \rangle = \frac{1}{4} \quad \langle h_2^T, h_3^T \rangle = \frac{1}{4}. \tag{48}$$

For bell-shaped type $\Upsilon_B(\mu)$ characterized by the two parameters (r and $1/p$) as in (28), we have

$$\Upsilon_B(\mu) = rh_1^B(\mu) + \frac{1}{p}h_2^B(\mu). \tag{49}$$

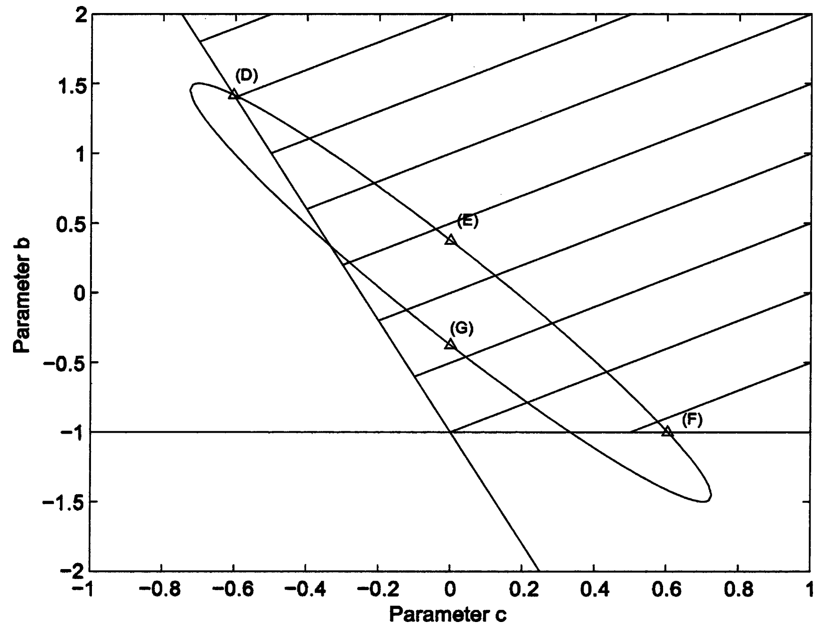

Fig. 11. Conclusions and well-defined (shaded) region for Example 2 with curved type orthogonal component.

The $h_i^B(\mu)$'s are given by

$$h_1^B(\mu) = 1, \qquad 0 \le \mu \le 2$$

$$h_2^B(\mu) = \begin{cases} -\frac{\cos^{-1}(2\mu-1)}{\pi}, & 0 \le \mu < 1 \\ 1 - \frac{\cos^{-1}(2(\mu-1)-1)}{\pi}, & 1 \le \mu \le 2 \end{cases}.$$

With $(g_{B_2} - g_{B_1}) = D_1 h_1^B + D_2 h_2^B$, conditions (42) and (43) become

$$r^2 \langle h_1^B, h_1^B \rangle + \left(\frac{1}{p}\right)^2 \langle h_2^B, h_2^B \rangle + 2\left(\frac{r}{p}\right) \langle h_1^B, h_2^B \rangle = \Delta^2 \tag{50}$$

$$rD_1 \langle h_1^B, h_1^B \rangle + \frac{D_2}{p} \langle h_2^B, h_2^B \rangle + \left(rD_2 + \frac{D_1}{p}\right) \langle h_1^B, h_2^B \rangle = 0. \tag{51}$$

The inner products are

$$\langle h_1^B, h_1^B \rangle = 1 \quad \langle h_2^B, h_2^B \rangle = \frac{(\pi^2 - 4)}{2\pi^2} \quad \langle h_1^B, h_2^B \rangle = 0. \tag{52}$$

Note that there can be a mix of different types of membership functions in an interpolation problem. For example, the consequents may be given in triangular membership functions with $(g_{B_2} - g_{B_1})$ thus expressed in $h_i^T(\mu)$'s, while an orthogonal component in curved membership function is desired. In this case, we need to calculate the mixed inner products of $\langle h_i^T, h_j^C \rangle$ as well. In any case, with the required inner products precomputed, the equality constraints defining the conclusion set are efficiently obtained by mere substitution of the value D_i's into the proper norm and orthogonal conditions. Conditions (30) and (31), (33) and (34), and (37) and (38) governing the conclusion sets for triangular, curved, and bell-shaped type orthogonal component in Example 2 are all obtained accordingly.

The second issue to address here concerns the extraction of unique conclusion. Depending on the number of parameters characterizing the set of conclusion, the problem may end up with more than one possible conclusions. As mentioned, additional objectives may be added to uniquely determine a conclusion in this case. Take the triangular membership function case in Example 2. Constraints (33) and (34) and the well-defined condition (36) yield the possible conclusion set of Figs. 9 and 10. We may further add a requirement that the basewidth of the conclusion be smallest possible, i.e., we desire a conclusion with the least degree of fuzziness. With $g_{B^*} = g_{B^\perp} + ah_1^T(\mu) + bh_2^T(\mu) + ch_3^T(\mu)$, the basewidth of the conclusion is dictated by the quantity $b + c$. We then have the following nonlinear programming problem to determine the unique conclusion:

$$\text{Min } b + c$$

subject to

$$a = -\left(\frac{3}{4}b + \frac{1}{4}c\right)$$
$$b^2 + 1.2bc + c^2 = 9.6\Delta^2$$
$$b \ge -1$$
$$c \ge -1.$$

Note that it is the norm condition that gives rise to the nonlinear equality in the problem. The remaining conditions are all linear for the three types of orthogonal components considered here. The above minimization problem can be efficiently solved by many commercially available software, such as *fmincon* in Matlab. The resulting conclusion is labeled (G) in Figs. 9 and 10. The minimum basewidth requirement is just one such possibility. Other objectives may be used to well define an unique conclusion under similar formulation.

The minimum basewidth conclusion using curved type orthogonal component in Example 2 can also be solved. The results are labeled (G) and shown in Figs. 11 and 12. The curved type case, however, is more complicated than the triangular type.

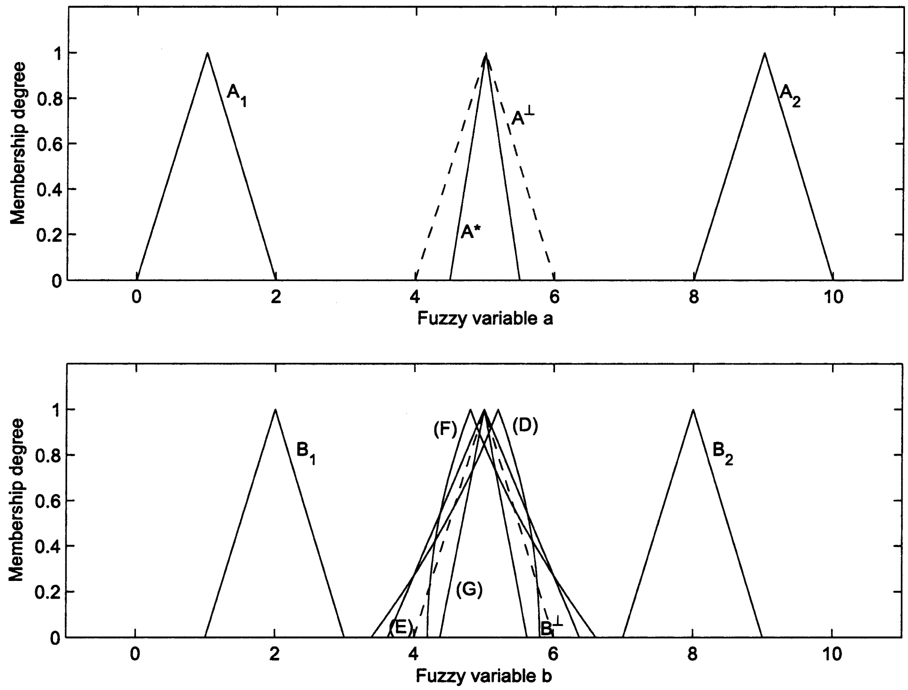

Fig. 12. $L_2[0, 2]$-interpolation for Example 2 with curved type orthogonal component.

Depending on the starting values assigned to the parameters, the command *fmincon* may converge to a local suboptimal only. This is due to the fact that the well-defined conditions subdivided the feasible region into disconnected sets. If the well-defined conditions were removed, *fmincon* would arrive at the global optimal efficiently. This points to the need in general case to check whether global minimization has indeed been reached using, for example, the KKT conditions [16].

VII. Spanning Set and Extensibility Function in $L_2[0, 2]$ Space

Other concepts introduced in [7] for the Cartesian interpolation of a general number of q given rules can be likewise extended to the $L_2[0, 2]$ space. They include the antecedent and consequent spanning sets. For a given set of rule antecedents $\{g_{A_1}, \ldots, g_{A_q}\}$ in $L_2[0, 2]$, the antecedent spanning set is defined as

$$\begin{aligned}\mathcal{G}_a = \{g_P \in L_2[0,2] | g_P = g_{A_r} &+ \alpha_1 (g_{A_1} - g_{A_r}) \\ &+ \alpha_2 (g_{A_2} - g_{A_r}) + \ldots + \alpha_{r-1} \left(g_{A_{r-1}} - g_{A_r}\right) \\ &+ \alpha_{r+1} \left(g_{A_{r+1}} - g_{A_r}\right) + \ldots \\ &+ \alpha_q \left(g_{A_q} - g_{A_r}\right)\}\end{aligned} \tag{53}$$

where α_i's are the real coefficients and g_{A_r} is the antecedent conveniently chosen as reference. Depending on the linear dependency of the terms involved, coefficients α_i may not be unique for a given g_P. The consequent spanning set can be similarly defined. The meaning of the spanning set is that its elements share the same characteristics. An observation lying within the antecedent spanning set embeds the same characteristics as the rule antecedents. Interpolation can hence be conducted based solely on the given rules, resulting in a conclusion lying inside the consequent spanning set. No additional information outside the rules is needed in this case.

Another concept in [7] that can be extended is the extensibility function. For a given j^{th} rule: $A_j \rightarrow B_j$, the extensibility function $E_j(g_{A^*}) = E_j(|g_{A^*} - g_{A_j}|)$ is a function of g_{A^*} measuring the degree of applicability of the jth rule at the location g_{A^*} not necessary equal to g_{A_j}. Physically, $E_j(g_{A^*})$ reflects how reliable the j^{th} rule can be applied at the neighboring points of the antecedent g_{A_j}. Extensibility functions can be different for different rules. Possible choices include

$$E_j(g_{A^*}) = \frac{1}{\beta \left|g_{A^*} - g_{A_j}\right|} \tag{54}$$

$$E_j(g_{A^*}) = e^{-\beta\left|g_{A^*} - g_{A_j}\right|} \tag{55}$$

$$E_j(g_{A^*}) = e^{-\beta\left|g_{A^*} - g_{A_j}\right|^2} \tag{56}$$

where β is a positive constant depending the validity distance of the rule involved, and $|.|$ is the defined $L_2[0, 2]$-norm. With the extensibility functions assigned, one may conduct interpolation for a given observation g_{A^*} using the weighted-sum average

$$g'_{B^*}(g_{A^*}) = \frac{\sum_{j=1}^{q} E_j(g_{A^*}) g_{B_j}}{\sum_{j=1}^{q} E_j(g_{A^*})}. \tag{57}$$

Equation (57) specifies that g'_{B^*} is a function of g_{A^*}. It can be proved that $g'_{B^*}(g_{A^*})$ always lies in the consequent spanning set. This is so even if g_{A^*} is not in the antecedent spanning set. Equation (57) actually maps the whole antecedent space into the consequent spanning set.

To facilitate a more general interpolation than (57), the work [7] included a scheme to produce a g_{B^*} not restricted to the consequent spanning set in the case that g_{A^*} is not in the antecedent spanning set. This scheme can also be generalized to $L_2[0, 2]$. The observation g_{A^*} can be decomposed into two components, one projected onto the antecedent spanning set, and the other orthogonal to it. For the component projected on the antecedent

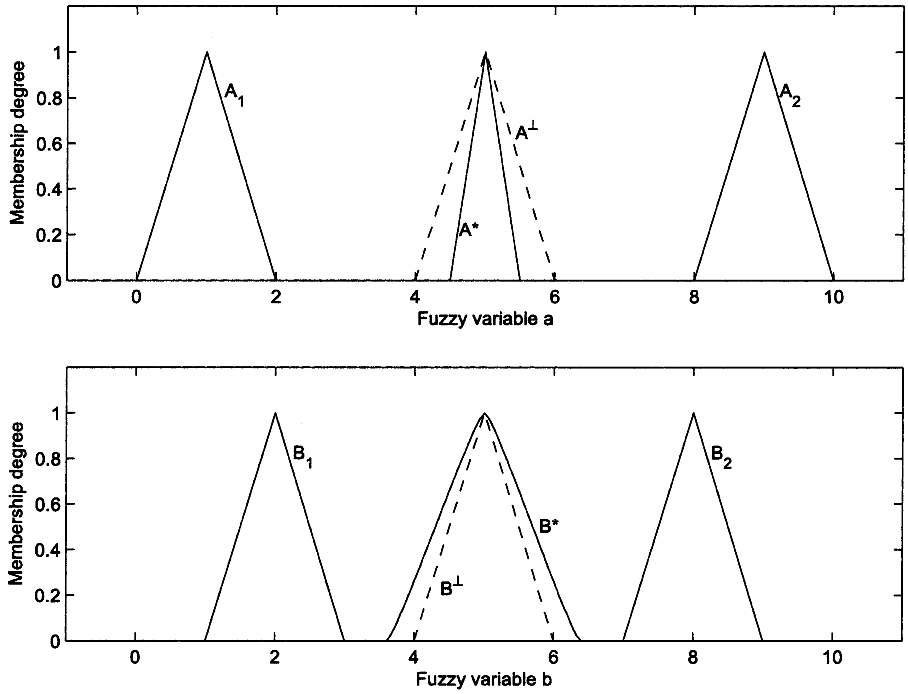

Fig. 13. $L_2[0,2]$-interpolation for Example 2 with bell-shaped orthogonal component.

spanning set, denoted as $g_{A^\perp}$, interpolation is conducted with (57) to yield $g_{B^\perp} \equiv g'_{B^*}(g_{A^\perp})$ inside the consequent spanning set.

The orthogonal component $(g_{A^*} - g_{A^\perp})$, however, should result in a component orthogonal to the consequent spanning set. Denote the interpolated conclusion as g_{B^*}. The orthogonal component to the consequent spanning set is then $(g_{B^*} - g_{B^\perp})$. Let the norm value of this component be $|g_{B^*} - g_{B^\perp}| = \Delta$. A scaling equation for Δ can be stated

$$\frac{|g_{A^*} - g_{A^\perp}|}{\mathcal{C}(\mathcal{G}_a)} = \frac{\Delta}{\mathcal{C}(\mathcal{G}_c)} \tag{58}$$

where $\mathcal{C}(\mathcal{G}_a)$ and $\mathcal{C}(\mathcal{G}_c)$ are certain measures to define the "core" values of the given set of antecedents and consequents. One possible definition of $\mathcal{C}(\mathcal{G}_a)$ is

$$\mathcal{C}(\mathcal{G}_a) = \frac{1}{q}\sum_{j=1}^{q} \left| g_{A_j} - g_{A^\perp} \right| \tag{59}$$

and similarly for $\mathcal{C}(\mathcal{G}_c)$. Equation (58) reduces to (19) of the similarity triangle method for $q = 2$ and the choice of (59). An example is now given for the $q = 3$ case.

Example 3: Consider the $L_2[0,2]$ interpolation of $q = 3$ rules, $A_i \rightarrow B_i$, $i = 1, 2, 3$. The corresponding membership functions g_{A_i}, g_{B_i}, and g_{A^*} in this case are bell-shaped type of the form

$$\Upsilon'_B(\mu) = \begin{cases} r - \frac{cos^{-1}(2\mu-1)}{p_1\pi}, & 0 \le \mu < 1 \\ r + \frac{1}{p_2}\left(1 - \frac{cos^{-1}(2(\mu-1)-1)}{\pi}\right), & 1 \le \mu \le 2 \end{cases}. \tag{60}$$

Comparing to $\Upsilon_B(\mu)$ before, $\Upsilon'_B(\mu)$ here allows the possibility of different base values to the left and the right side of the membership function, namely, $1/p_1$ and $1/p_2$. The parameter r remains the peak location. Given two membership functions of such kind

$$g_x(\mu) = \begin{cases} r_x + \frac{-cos^{-1}(2\mu-1)}{p_{x1}\pi}, & 0 \le \mu < 1 \\ r_x + \frac{1}{p_{x2}}\left(1 - \frac{cos^{-1}(2(\mu-1)-1)}{\pi}\right), & 1 \le \mu \le 2 \end{cases}$$

$$g_y(\mu) = \begin{cases} r_y + \frac{-cos^{-1}(2\mu-1)}{p_{y1}\pi}, & 0 \le \mu < 1 \\ r_y + \frac{1}{p_{y2}}\left(1 - \frac{cos^{-1}(2(\mu-1)-1)}{\pi}\right), & 1 \le \mu \le 2 \end{cases}$$

their inner product is determined as

$$\langle g_x, g_y \rangle = r_x r_y + \frac{1}{2}\left(\frac{1}{p_{x1}p_{y1}} + \frac{1}{p_{x2}p_{y2}}\right)\frac{(\pi^2 - 4)}{2\pi^2}.$$

Fig. 14 shows the corresponding membership functions g_{A_i}, g_{A^*}, and g_{B_i} of the present example. By design, the consequent spanning set is parameterized by one variable, thereby allowing for a two-variable parameterization of the orthogonal space. The present example is same as the numerical example in [7, Sec. VI], except that there triangular membership functions and Cartesian interpolation were applied.

Projecting observation g_{A^*} onto the antecedent spanning set, we obtain

$$g_{A^\perp} = g_{A_1} + \alpha_1\left(g_{A_2} - g_{A_1}\right) + \alpha_2\left(g_{A_3} - g_{A_1}\right) \tag{61}$$

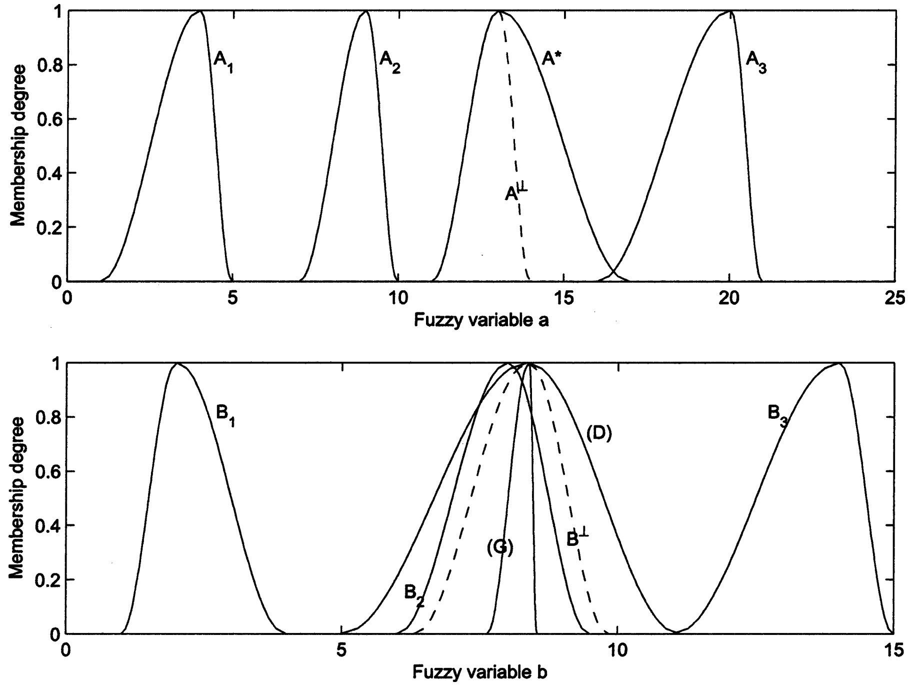

Fig. 14. $L_2[0,2]$-interpolation for Example 3.

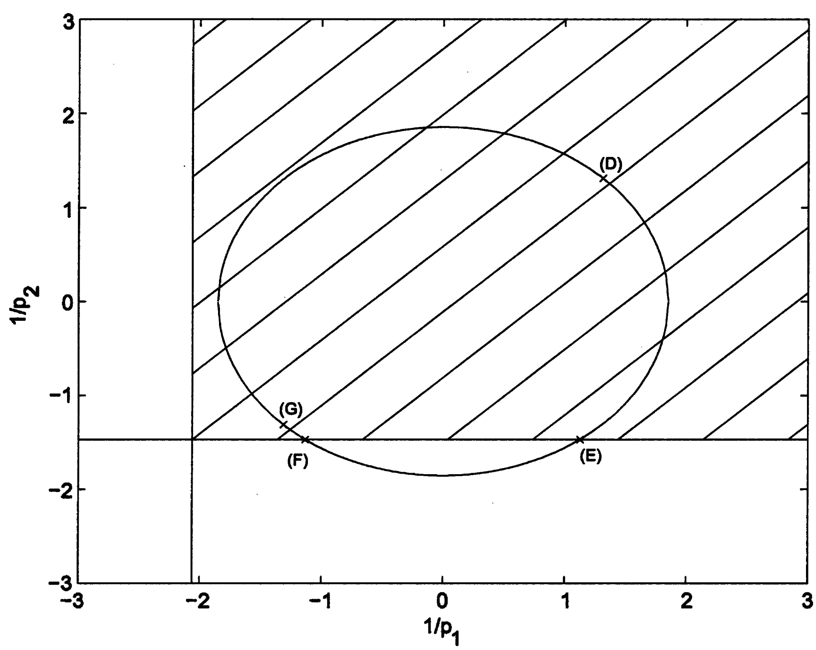

Fig. 15. Conclusions and well-defined (shaded) region for Example 3 with $L_2[0,2]$ interpolation.

with $\alpha_1 = 1.1905$ and $\alpha_2 = 0.1905$. The image of $g_{A^\perp}$ in the consequent spanning set by (57) is

$$g_{B^\perp} = \begin{cases} 8.3696 - \frac{1}{0.4851}\frac{cos^{-1}(2\mu-1)}{\pi}, & 0 \le \mu < 1 \\ 8.3696 \\ \quad +\frac{1}{0.6806}\left(1 - \frac{cos^{-1}(2(\mu-1)-1)}{\pi}\right), & 1 \le \mu \le 2 \end{cases} \tag{62}$$

where extensibility function (54) with $\beta = 1$ has been assigned to all rules. Continuing, we obtain $\mathcal{C}(\mathcal{G}_a) = 6.6835$, $\mathcal{C}(\mathcal{G}_c) = 4.1338$, $|g_{A^*} - g_{A^\perp}| = 1.1568$ and hence, $\Delta = 0.7155$. Now, let the orthogonal component be also in the form of (60). All quantities in this example, g_{B^*} included, are hence bell-shaped membership function of the type $\Upsilon'_B(\mu)$. The norm and orthogonal conditions then yield

$$r^2 + 0.1487\left(\frac{1}{p_1^2} + \frac{1}{p_2^2}\right) = 0.5119 \tag{63}$$

$$r + 0.0248\frac{1}{p_1} - 0.0124\frac{1}{p_2} = 0 \tag{64}$$

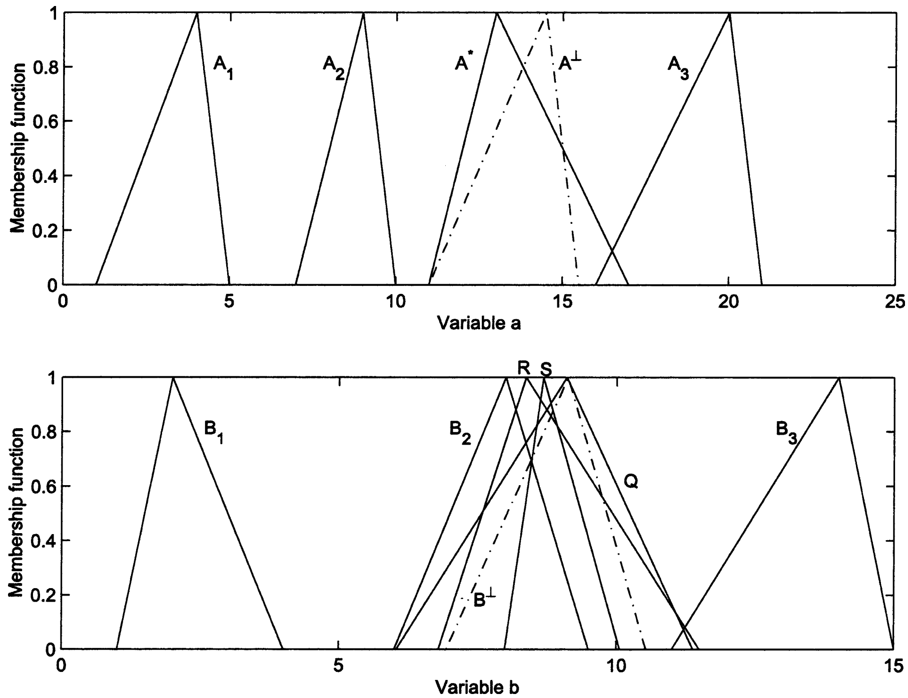

Fig. 16. Would-be interpolation results of Example 3 with Cartesian interpolation.

for the defining parameters of the orthogonal component. The well-defined conditions (23) are

$$\frac{1}{p_1} \geq -2.0616 \tag{65}$$

$$\frac{1}{p_2} \geq -1.4692. \tag{66}$$

Fig. 15 shows the possible solution superimposed on the well-defined (shaded) region in the $((1/p_1), (1/p_2))$ plane. Any point on the ellipse inside the well-defined region would correspond to a valid solution. Specifically. the points (E) and (F) give rise to solutions having a nonzero left side of the bell-shape only. Points (D) and (G) correspond to conclusions having, respectively, the maximum and minimum values in total basewidth. Membership functions for conclusions (D) and (G) are included in Fig. 14. For comparison, Fig. 16 shows the analogous results from [7] using triangular membership functions and Cartesian interpolation. The two sets of interpolated results are consistent to each other.

We have so far considered the interpolation of fuzzy rules with single antecedent in $L_2[0, 2]$. Possible extension to the multiple antecedent case can also be conducted as in [7] using vector function space. For fuzzy rules of the form: If $U_i, V_j, \ldots, X_k \rightarrow B_{ij\ldots k}$, one may form an enlarged antecedent space with elements of

$$\vec{g}_{A_s} = \begin{bmatrix} g_{U_i} \\ g_{V_j} \\ \vdots \\ g_{X_k} \end{bmatrix} \tag{67}$$

where $g_{U_i}, g_{V_j}, \cdots, g_{X_k}$ are membership functions of the individual antecedent under $L_2[0, 2]$ representation. The fuzzy rules can thus be taken as mappings from the antecedent space of a vector function space to a consequent space of a $L_2[0, 2]$ space. The inner product and the norm for this vector function space can be defined appropriately to tradeoff the involved complexity and conservativeness. This will be a topic of future study.

VIII. Conclusion

Interpolation is an important subject for reducing dense rule bases as well for deriving useful information from sparse ones. A previous approach to facilitate interpolation via representing fuzzy rules as mappings between finite dimensional Cartesian spaces has been proposed. The present work is an indepth study to expand the previous approach. Membership functions are now represented as elements in the space of square integrable functions and fuzzy rules as mappings between two such spaces. The new representation accommodates an extended class of fuzzy rules with membership functions satisfying two monotonicity conditions. They include smooth membership functions such as the popular bell-shaped membership functions, which were not possible before with the Cartesian representation. Requirement on well-defined conclusion now translates to become a constraint on the nonnegative slope of the function space representation. The new approach enables different types of membership functions, e.g., triangular, trapezoidal, and bell-shaped, to be treated under an unified framework, allowing possible cross-type interpolation studies. The new approach is more computational intensive. With the definition and precomputation of proper basis functions and inner products, however, computational efficiency can be enhanced during actual processing. Additional objective(s) incorporated to effectuate unique interpolated conclusion may also be formulated as a nonlinear programming problem. The present work demonstrates the application of interpolation based on geometric concepts such as similarity triangles and core values. Other methods such as treating the mappings as samplings of a nonlinear transformation may be attempted. Additional membership function types besides the three considered here may also be studied. More generally, the present approach allows a rigorous study and comparison of fuzzy inference subject to different or mixed types of membership functions. The results here suggest that bell-shaped membership function case can be treated by viewing the finite number of defining parameters as "coordinates" in a finite dimensional space equipped with special inner product. This will be another direction to pursue. The ultimate goal of our research is the application of the wealth of mathematical results in scalar and vector function space mapping and transformation to the areas of fuzzy inference, interpolation, and extraction.

REFERENCES

[1] L. T. Kóczy and K. Hirota, "Approximate reasoning by linear rule interpolation and general approximation," *Int. J. Approx. Reason.*, vol. 9, pp. 197–225, 1993.

[2] ——, "Interpolative reasoning with insufficient evidence in sparse fuzzy rule bases," *Inform. Sci.*, vol. 71, pp. 169–201, 1993.

[3] P. Baranyi, T. D. Gedeon, and L. T. Kóczy, "A general interpolation technique in fuzzy rule bases with arbitrary membership functions," in *Proc. IEEE Int. Conf. Systems, Man, and Cybernetics*, ch. China, Beijing, 1996, pp. 510–515.

[4] T. D. Gedeon and L. T. Kóczy, "Conservation of fuzziness in rule interpolation," in *Proc. Intelligent Technologies Int. Symp. on New Trends in the Control of Large Scale Systems*, vol. 1, Herlany, Slovakia, 1996, pp. 13–19.

[5] L. T. Kóczy, K. Hirota, and T. D. Gedeon, "Fuzzy rule interpolation by the conservation of relative fuzziness," Hirota Lab., Dept. Intell. Comput. Syst. Sci., Tokyo Inst. Technol., Yokohama, Japan, Tech. Rep. TR 97-2, 1997.

[6] Y. Yam and L. T. Kóczy, "Cartesian representation for fuzzy interpolation," in *Proc. 37th Conf. Decision and Control*, Tampa, FL, Dec. 16–18, 1998, pp. 2936–2937.

[7] ——, "Representing membership functions as points in high dimensional spaces for fuzzy interpolation and extrapolation," *IEEE Trans. Fuzzy Syst.*, vol. 8, no. 6, pp. 761–772, Dec. 2000.

[8] Y. Yam, M. L. Wong, and P. Baranyi, "Interpolation as mapping between cartesian spaces," in *Proc. Int. Conf. Intelligent Technologies (InTech'2000)*, Bangkok, Thailand, 2000, pp. 450–459.

[9] Y. Yam, P. Baranyi, D. Tikk, and L. T. Kóczy, "Eliminating the abnormality problem of α-cut based fuzzy interpolation," in *Proc. 8th Int. Fuzzy Systems Association World Congr. (IFSA'99)*, Taipei, Taiwan, Aug. 17–20, 1999, pp. 762–766.

[10] Y. Yam, V. Kreinovich, and H. T. Hung, "Extracting fuzzy sparse rule base by cartesian representation and clustering," in *Proc. 2000 IEEE Int. Conf. Systems, Man, and Cybernetics*, Nashville, TN, pp. 3778–3783.

[11] M. L. Wong, Y. Yam, and P. Baranyi, "Representing membership functions as elements in function space," in *Proc. Amer. Control Conf.*, vol. 3, 2001, pp. 1922–1927.

[12] J. N. Reddy, *Applied Functional Analysis and Variational Methods in Engineering*. New York: McGraw-Hill, 1986.

[13] W. J. Wang and C. H. Chiu, "Entropy change in extension principle," *Fuzzy Sets Syst.*, vol. 103, pp. 153–162, 1999.

[14] P. Baranyi, D. Tikk, Y. Yam, and L. T. Kóczy, "Investigation of a new α-cut based fuzzy interpolation method," Dept. Mech. Automat. Eng., Chinese Univ. Hong Kong, Tech. Rep. CUHK-MAE-99-06, 1999.

[15] M. L. Wong, "Further investigations of geometric representation approach to fuzzy inference and interpolation," M.Phil. thesis, Dept. Mech. Automat. Eng., Chinese Univ. Hong Kong, 2002.

[16] A. D. Belegundu and T. R. Chandrupatla, *Optimization Concepts and Applications in Engineering*. Upper Saddle River, NJ: Prentice-Hall, Inc., 1999.

Yeung Yam (M'92–SM'01) received the B.S. and M.S. degrees in physics from the Chinese University of Hong Kong and the University of Akron, Akron, OH, respectively, in 1975 and 1977, and the M.S. and Sc.D. degrees in aeronautics and astronautics from the Massachusetts Institute of Technology, Cambridge, in 1979 and 1983, respectively.

He joined the Chinese University of Hong Kong in 1992, and is currently the Chairman of the Department of Automation and Computer-Aided Engineering. Before joining the university, he was with the Control Analysis Research Group of the Guidance and Control Section at Jet Propulsion Laboratory, Pasadena, CA, USA. He is a senior member of IEEE. His research interests include intelligent control, fuzzy approximation, system identification, dynamics modeling and analysis. His has published over 100 technical papers in various areas of his fields.

Man Lung Wong received the B.Eng and M.Phil. degrees in mechanical and automation engineering from the Chinese University of Hong Kong, in 1999 and 2002, respectively.

His research interest is in fuzzy interpolation and representation and his thesis is entitled "Further investigations of geometric representation approach to fuzzy inference and interpolation." He is now with Opera Audio as an audio engineer.

Péter Baranyi was born in Hungary in 1970. He received the M.Sc. degree in electrical engineering, the M.Sc. degree in education of engineering sciences, and the Ph.D. degree, all from the Technical University of Budapest, Budapest, Hungary, in 1994, 1995, and 1999, respectively.

He has had research positions at the Chinese University of Hong Kong (1996 and 1998), the University of New South Wales, Australia (1997), the CNRS LAAS Institute, Toulouse, France (1996), the Technical University of Buisberg, Germany (1997), Gifu Research Institute, Japan (2000–2001), the University of Hull, U.K., among others. His research interest includes nonlinear control and fuzzy interpolation.

Dr. Baranyi received the Youth Prize of the Hungarian Academy of Sciences (2000), the International Dennis Gabor Award (2000), the STA Award from the Japan Science and Technology Corporation, Department of International Affairs (2001), the Young Technological Innovator of the Year Prize (2002), and the Young Scientist Prize from Samsung (2003). He is the Vice President of the Hungarian Society of the International Fuzzy Systems Association (IFSA).